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ChE 210 Exam #1 1 October 2025

To receive full credit, you must show all of your work.

Species	Molar Mass
CP ($Ca_3(PO_4)_2$)	310.18
SA (H_2SO_4)	98.08
PA (H_3PO_4)	97.994
CS ($CaSO_4$)	136.14
W (H_2O)	18.02

(1) (50 pts.) Phosphoric acid (PA) can be produced by reacting calcium phosphate (CP) with a stoichiometric quantity of sulfuric acid (SA). Calcium sulfate (CS) is also a byproduct of this reaction. Two streams enter the reactor: (**Stream 1**) has a species mass flow rate of 10,000 kgCP/hr of CP which is suspended in an unknown amount of water (W) and (**Stream 2**) a 95 wt% aqueous solution of SA. There are two exit streams: (**Stream 3**) a 40 wt% aqueous solution of PA and (**Stream 4**) a solid of composed of CS and water which has a mole ratio of 2 moles of water for each mole of CS.

(a) Write a balanced chemical reaction equation for this reaction. (5 pts.)

$$\underset{CP}{1\,Ca_3(PO_4)_2} + \underset{SA}{3H_2SO_4} \longrightarrow \underset{PA}{2H_3PO_4} + \underset{CS}{3CaSO_4}$$

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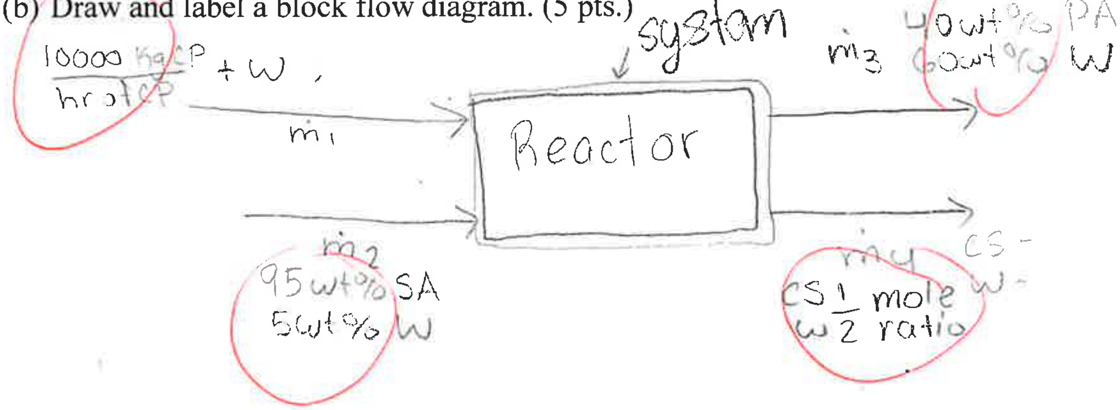
(b) Draw and label a block flow diagram. (5 pts.)



5

(c) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. In the table below, give values for 4a, 5a, 5b, 5c, and 5d along with a brief explanation for the value you give. Notice that the value for 4b is given with an explanation. Compute the DOF. (10 pts.)

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	Number	Explanation
4a. Stream Variables	8 ✓	4 streams × 2 variable in each.
4b. System Variable	0	No reactants in product streams. The reaction is complete.
Total Independent Variables	8 9	
5a. Specified Flows	1	BASIS of 10,000 kg CP / hr of CP
5b. Specified Stream Compositions	2 3	95 wt % SA, 40 wt % PA
5c. Specified System Performances	0	No system performance
5d. Material Balance Equations	5	5 Compounds for which balances can be written
Total Independent Equations	8 9	
DOF	0 1	Well Posed

[Problem 2 continued]

(c) (30 pts.) Calculate the species molar flow rates of the ammonia leaving the system, that is, A2 and A4. This calculation is likely to require solving a quadratic equation.

$\dot{n}_1 + \dot{n}_3 = \dot{n}_2 + \dot{n}_4$ ✓ $37.174 \frac{kmol}{hr} + 55.556 \frac{kmol}{hr} = \dot{n}_2 + \dot{n}_4$ ✓

$92.730 = \dot{n}_2 + \dot{n}_4$ ✓ and $(X_{2A}) = 2(X_{4A})$

$92.730 = X_{N_2}(\dot{n}_2) + X_{A_2}(\dot{n}_2) + X_{W4}(\dot{n}_4) + X_{A4}(\dot{n}_4)$ ok

$3.717 = X_{A2}(\dot{n}_2) + X_{A4}(\dot{n}_4)$ ✓

$3.717 = X_{A2}(\dot{n}_2) + 2X_{A2}(\dot{n}_4)$ ✓ $2x(2) + x(3)$

$= (2\dot{n}_4 + \dot{n}_2) X_{A2}$

need more balance eqns.

$\frac{3.717}{92.730} = A_2$

10

$X_{A2} = 0.04008$ $X_{A4} = 0.0801$

$\frac{3.717}{3} = \frac{3X_{A2}}{3}$ $X_{A2} =$

$N_2 =$ 33.457 ✓ kmolN/hr $A_2 =$ ~~1.239~~ kmolA/hr $W_4 =$ 55.556 ✓ kmolW/hr $A_4 =$ ~~2.478~~ kmolA/hr

(d) (5 pts.) What is the percent recovery of the ammonia, that is, what percentage of the ammonia entering the separator is removed with the water stream?

0

25

$\left|\frac{f - i}{i}\right| \times 100$

-33.33%

LOST

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(2) **(50 pts.)** A gas mixture of nitrogen N_2 (N) and ammonia NH_3 (A) enters the bottom of a separator as **Stream 1** with mass flow rate $\dot{m}_1 = 1000$ kg/hr and a mole fraction of the ammonia $x_{A1} = 0.10$. **Stream 2** exits the top of the separator containing a gas mixture of nitrogen and ammonia. Pure liquid water H_2O (W) enters the top of the same separator as **Stream 3** at a mass flow rate of $\dot{m}_3 = 1000$ kg/hr and a liquid mixture of water and ammonia exits the bottom as **Stream 4**. The nitrogen and water do not mix during the process. The mole fractions of the ammonia in the two exiting streams are known to be related by $x_{2A} = 2x_{4A}$. Assume steady state. Molecular Weights: $M_A = 17$ kg/kmol for ammonia, $M_N = 28$ kg/kmol for nitrogen, and $M_W = 18$ kg/kmol for water.

$\dot{m}_3$ 1000 kg/hr, $\omega_{3}=1$; $x_{A2}=?$ $x_{N2}=?$ $\dot{m}_2$; $x_{W4}=?$ $x_{A4}=?$; $\dot{m}_1$ 1000 kg/hr, $x_{A1}=0.10$, $x_{N1}=0.90$

(a) (5 pts.) Calculate the mass fraction of A in Stream 1, that is w_{1A}.

$$W_{1A} = \frac{\text{mass}_{1A}}{\text{total mass}} = \frac{(100\,\text{kg mol A})\left(\frac{17\,\text{kg}}{\text{kmol A}}\right)}{100\,\text{kmol}\,\frac{\text{kg}}{\text{kg mol}}\,A + 900\,\text{kg mol}\left(28\,\frac{\text{kg}}{\text{mol}}\,N\right)}$$

$W_{1A} = 0.063$

$\boxed{63.197\ \frac{\text{kg A}}{\text{hr}}}$

$\boxed{936.802\ \text{kg N/hr}}$

$w_{1A} =$ 0.063 kgA/kg

(b) (10 pts.) Calculate the species molar flow rates for the inlet ammonia A_1, nitrogen N_1, and water W_3.

$$\frac{0.063\,\text{kg A}}{\text{kg}} \times \frac{1000\,\text{kg}}{\text{hr}} = \frac{63.197\,\text{kg A}}{\text{hr}} \cdot \frac{1\,\text{kmol A}}{17\,\text{kg A}} = 3.717\,\frac{\text{kmol A}}{\text{hr}}$$

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$$\frac{1000\,\text{kg}}{\text{hr}} - \frac{63.197\,\text{kg A}}{\text{hr}} = \frac{936.802\,\text{kg N}}{\text{hr}} \cdot \frac{1\,\text{kmol N}}{28\,\text{kg N}} = 33.4572\,\frac{\text{kmol N}}{\text{hr}}$$

$$\frac{1000\,\text{kg}}{\text{hr}} \times \frac{1\,\text{kg W}}{\text{kg}} = \frac{1000\,\text{kg W}}{\text{hr}} \times \frac{1\,\text{kmol W}}{18\,\text{kg}} = 55.56\,\frac{\text{kmol W}}{\text{hr}}$$

10

$A_1 =$ 3.717 kmolA/hr $N_1 =$ 33.457 kmolN/hr $W_3 =$ 55.556 kmolW/hr

[Problem 1 continued] $\dot{n}$

(d) Calculate the species molar flow rates for the four non-water components in the four streams in kmol/hr. That is CP_1, SA_2, PA_3, and CS_4 where the subscripts are stream numbers. Also, write a balance on water and calculate W_1, the water entering the process with CP_1. Show all your calculations and write your final answers in the spaces provided below. (30 pts.)

BASIS: $\frac{10000 \text{ kg CP}}{1 \text{ hr}} \times \frac{1 \text{ Kmol CP}}{0.31018 \text{ kg CP}} = \frac{32239.3449 \text{ kmol CP}}{\text{hr}} = \dot{n}_1 - w$

$\frac{32239.3449 \text{ KmolCP}}{\text{hr}} \times \frac{3 \text{ Kmol SA}}{1 \text{ Kmol CP}} = \frac{96718.0347 \text{ kmol SA}}{\text{hr}} = \dot{n}_2 - 0.05\dot{n}_2$

$\frac{32239.3449 \text{ kmol CP}}{\text{hr}} \times \frac{2 \text{ Kmol PA}}{1 \text{ KmolCP}} = \frac{64478.69 \text{ Kmol PA}}{\text{hr}} = \dot{n}_3 - 0.6\dot{n}_3$

$\frac{32239.3449 \text{ Kmol CP}}{\text{hr}} \times \frac{3 \text{ Kmol CS}}{1 \text{ Kmol CP}} = \frac{96718.0347 \text{ Kmol CS}}{\text{hr}} = \dot{n}_4 - (?)$

$\frac{96718.0347 \text{ Kmol CS}}{\text{hr}} \times \frac{2 \text{ Kmol W}}{\text{Kmole CS}} = \frac{193436.0694 \text{ kmol W}}{\text{hr}}$

$\dot{n}_1 + \dot{n}_2 \cancel{\times} \dot{n}_3 + \dot{n}_4$ No!

$\dot{n}_1 = CP + \boxed{W}$

$\dot{n}_2 = W_{SA}(\dot{n}_2) + W_W(\dot{n}_2)$

$\dot{n}_3 = W_{PA}(\dot{n}_3) + W_W(\dot{n}_3)$

$\dot{n}_4 = CS + W$ [1 mol CS : 2 mol W]

$\dot{n}_2 = 96718.034 + 0.05(\dot{n}_2)$

$0.95(\dot{n}_2) = 96718.034$

$\dot{n}_2 = \frac{101,808.4576 \text{ kmol}}{\text{hr}}$

$(\dot{n}_1 - CP) + (\dot{n}_2 - 0.95(\dot{n}_2)) = (\dot{n}_3 - 0.40(\dot{n}_3)) + (\dot{n}_4 - CS)$

$W_1 + W_2 = W_3 + W_4$

$\dot{n}_3 = \frac{161,196.7245 \text{ kmol}}{\text{hr}}$

$\dot{n}_4 = \frac{290154.1041 \text{ kmol}}{\text{hr}}$

22

42

1	42
2	25
	67

Species molar flow rate. Subscripts indicate stream numbers.

CP_1 (kmol/hr)	SA_2 (kmol/hr)	PA_3 (kmol/hr)	CS_4 (kmol/hr))	W_1 (kmol/hr)
32239.34	96718.03	64478.69	96718.03	317,303.03